

Unit 1.5 (Study Guide) — BASIC GRAPHS Self Study

Unit 1.6 (Study Guide)
Section 1.3 (Stewart)

New functions from Old functions (BASIC)

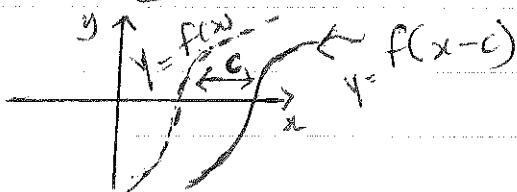
Suppose $c > 0$

VERTICAL AND HORIZONTAL SHIFTS

- To graph $y = f(x) + c$, shift the graph of $y = f(x)$ upward by c units
- To graph $y = f(x) - c$, shift the graph of $y = f(x)$ downward by c units
- To graph $y = f(x - c)$, shift the graph of $y = f(x)$ to the right by c units
- To graph $y = f(x + c)$, shift the graph of $y = f(x)$ to the left by c units

When sketching shifted graphs we dotted lines to indicate the shifts

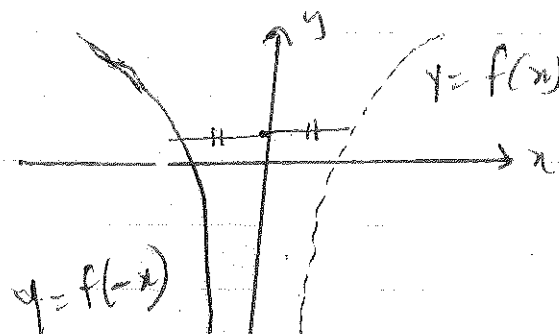
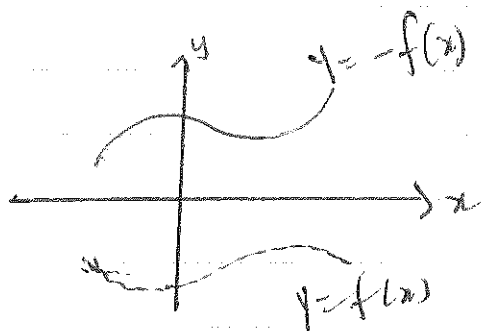
Also, try by all means to indicate x - and y -intercepts



REFLECTING GRAPHS

To graph $y = -f(x)$, reflect the graph of $y = f(x)$ in the x -axis

To graph $y = f(-x)$, reflect the graph of $y = f(x)$ in the y -axis.



BASIC FUNCTION/
OLD FUNCTION/

NEW FUNCTION

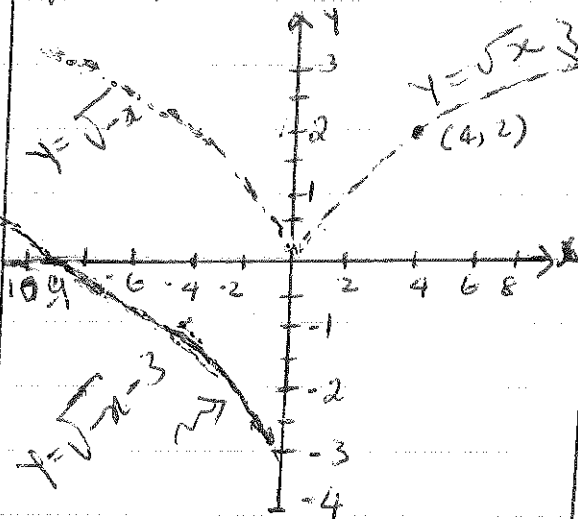
Transformation

1. $y = \sqrt{x}$



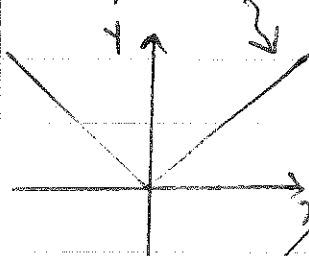
Square Root
function

$y = \sqrt{-x} - 3$

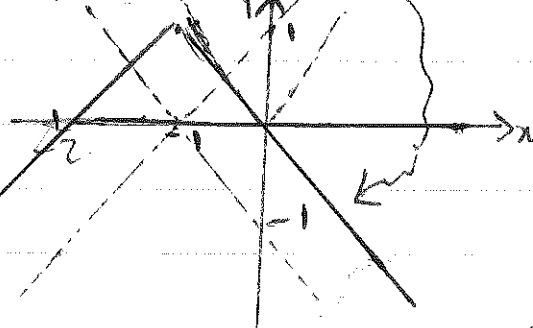


$y = \sqrt{x}$ is
Reflected in
the y -axis
and shifted
3 units
downward

2. $y = |x|$



$y = -|x+1| + 1$



$y = |x|$ is shifted
1 unit to the
left, then reflected
in the y -axis, then
shifted up by
1 unit.

Vertical stretching and shrinking of graphs

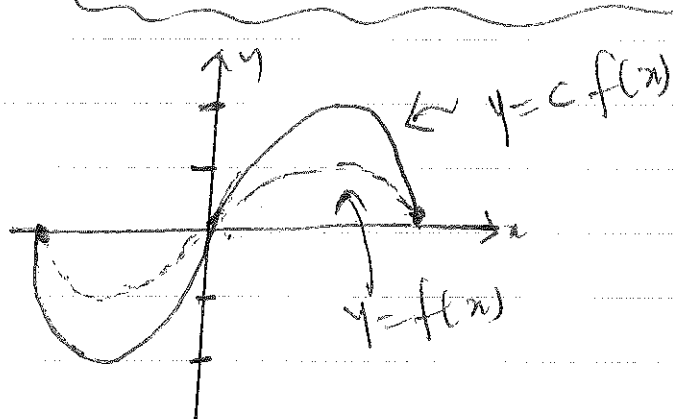
(*) To graph $y = c f(x)$:

- if $c > 1$, stretch the graph of $y = f(x)$ vertically by a factor of c

Multiplying by a number, larger than one, on the outside of a function causes the graph to be stretched or narrowed by a factor of c

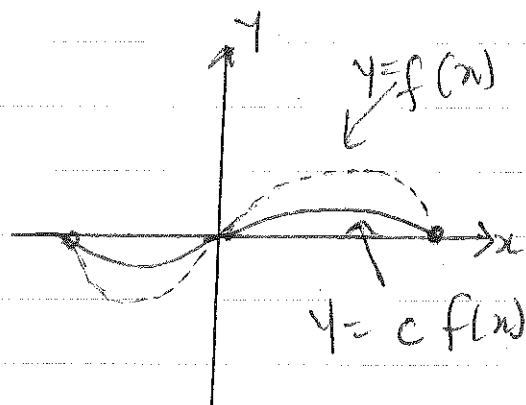
- if $0 < c < 1$, shrink the graph of $y = f(x)$ vertically by a factor of c

Multiplying by a number, between zero and one on the outside of a function causes the graph to shrink or widen by a factor of c



$c > 1$, stretch

VERTICALLY

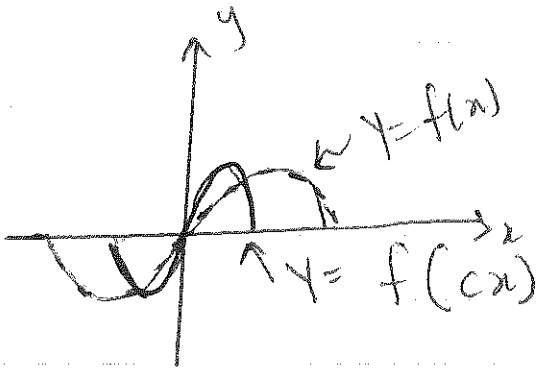


$0 < c < 1$
shrink

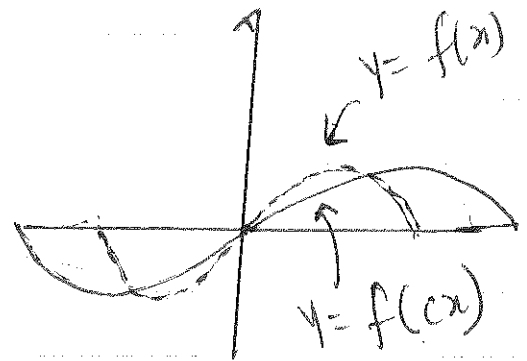
HORIZONTAL SHRINKING AND STRETCHING OF GRAPHS

(*) To graph $y = f(cx)$:

- If $c > 1$, shrink the graph of $y = f(x)$ horizontally by a factor of $\frac{1}{c}$
- If $0 < c < 1$, stretch the graph of $y = f(x)$ horizontally by a factor of $\frac{1}{c}$.



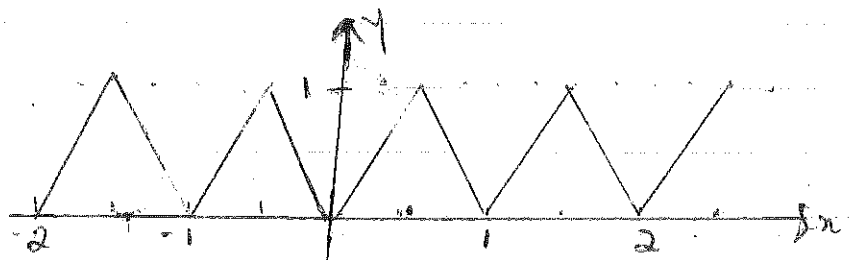
$c > 1$ Horizontal shrink.



$0 < c < 1$ stretch.

Example

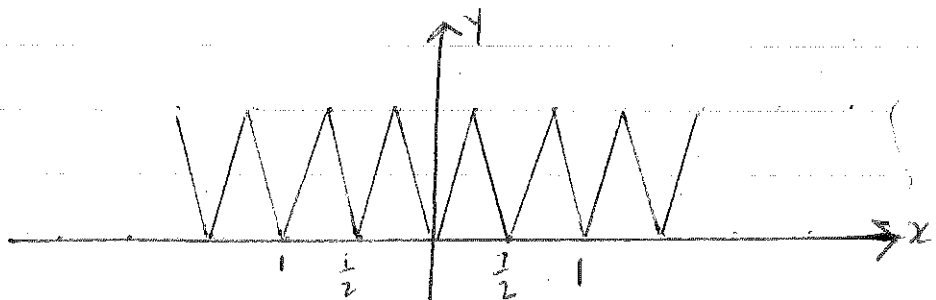
$y = f(x)$



$y = f(2x)$

Horizontal shrink

$c > 1$



Example: First Note!

$$(k > 0)$$

$y = a \sin kx$ and $y = a \cos kx$
have amplitude $|a|$ and period $\frac{2\pi}{k}$

$\Rightarrow$ An appropriate interval on which to graph one complete period would be $\frac{2\pi}{k}$ (instead of 2π)

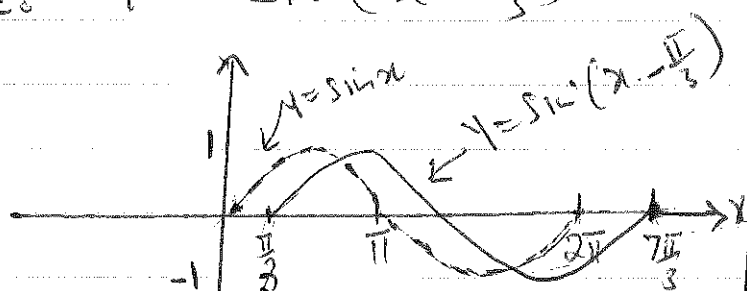
$$y = a \sin k(x-b) \text{ and } y = a \cos k(x-b)$$

$(k > 0)$

have amplitude $|a|$, period $\frac{2\pi}{k}$ and
phase shift b (shift to the right b units)

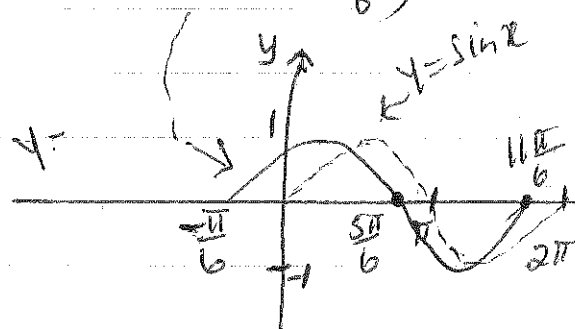
An appropriate interval on which to graph
one complete period is $[b, b + \frac{2\pi}{k}]$

1. $y = \sin(x - \frac{\pi}{3})$



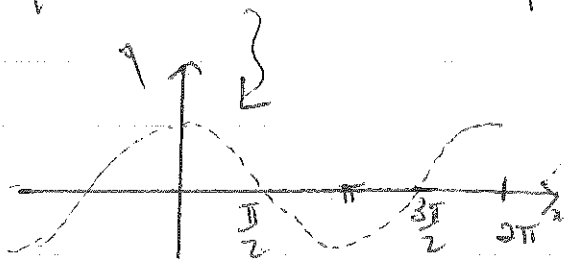
(Phase shift to the right)

$$y = \sin(x + \frac{\pi}{6})$$



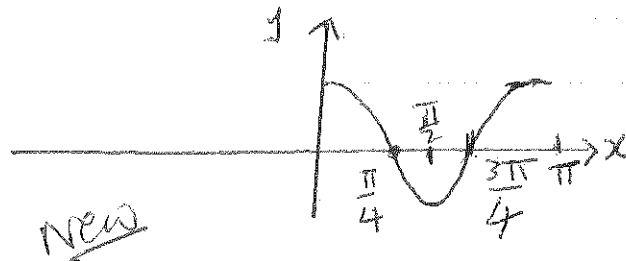
Examples :

$$y = \cos x$$



Basic
(old)

$$y = \cos 2x$$

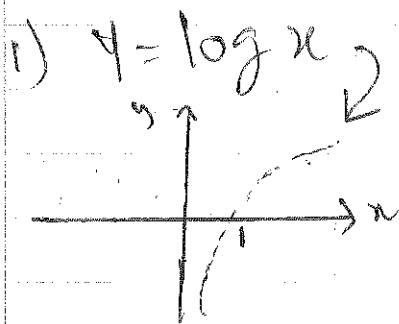


new

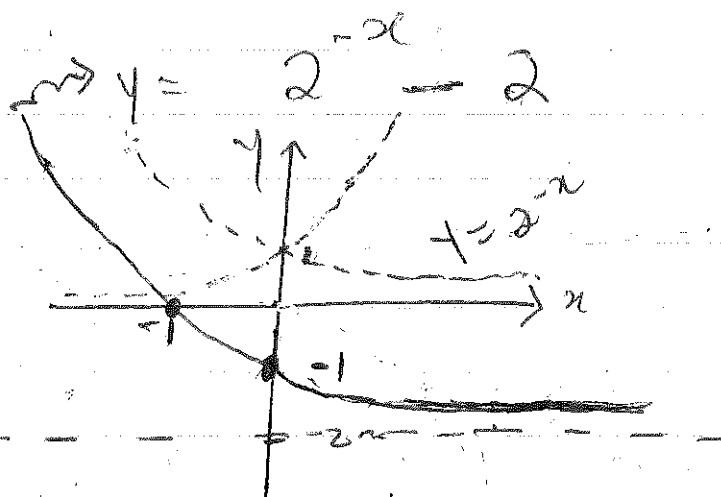
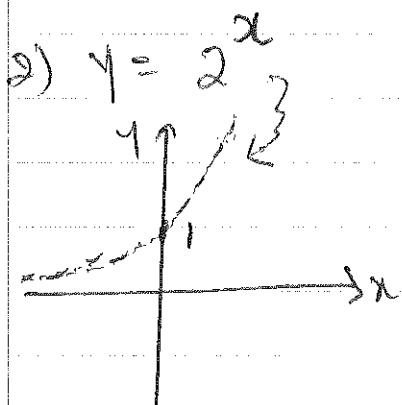
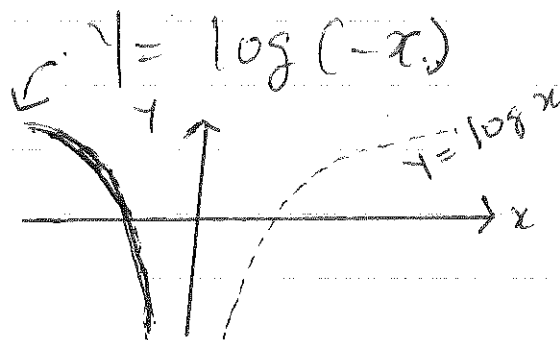
$$\text{period} = \frac{2\pi}{2} = \pi$$

Complete cycle is π (instead of 2π).

old graphs



New graphs



Indicate x and y intercepts.

COMBINATION OF FUNCTIONS

Let f and g be two functions with domains A and B

We can combine two functions f and g by : $f+g$, $f-g$, fg , f/g ($g \neq 0$)

defined as follows :

$$(f+g)(x) = f(x) + g(x) \quad \text{Domain: } A \cap B$$

$$(f-g)(x) = f(x) - g(x) \quad \text{" "}$$

$$(fg)(x) = f(x)g(x) \quad \text{" "}$$

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)} \quad \text{Domain: } \left\{x \in A \cap B \mid g(x) \neq 0\right\}$$

Example: let $f(x) = \frac{1}{x-2}$, $g(x) = \sqrt{x}$

$$a) \left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)} = \frac{1}{x-2} \div \sqrt{x}$$

$$= \frac{1}{x-2} \times \frac{1}{\sqrt{x}}$$

$$= \frac{1}{\sqrt{x}(x-2)}$$

Domain of $\frac{f}{g}$ is $\{x \mid x > 0 \text{ and } x \neq 2\}$

$$= (0, 2) \cup (2, \infty)$$

$$b) (f+g)(x) = \frac{1}{x-2} + \sqrt{x}$$

Domain is $\{x \mid x \geq 0 \text{ and } x \neq 2\} = [0, 2) \cup (2, \infty)$

Two functions f and g , ^(with domains A and B) can also be composed, defined by

$$(f \circ g)(x) = f(g(x))$$

called f composed with g .

Example:

If $f(x) = \sqrt{x}$ and $g(x) = \sqrt{2-x}$, find

- a) $(f \circ g)(x)$ b) $(g \circ f)(x)$ c) $(f \circ f)(x)$ d) $g \circ g(x)$

and their domains

Solution:

$$\begin{aligned} \text{a) } (f \circ g)(x) &= f(g(x)) \\ &= f(\sqrt{2-x}) \\ &= \sqrt{\sqrt{2-x}} \end{aligned}$$

by definition

$$= \sqrt[4]{2-x}$$

Domain of $f \circ g$ is $\{x \mid 2-x \geq 0\}$

$$= \{x \mid x \leq 2\}$$

$$= (-\infty, 2]$$

$$\begin{aligned} \text{b) } (g \circ f)(x) &= g(f(x)) = g(\sqrt{x}) \\ &= \sqrt{2 - \sqrt{x}} \end{aligned}$$

Example: If $(g \circ f)(x) = 2^{x^2}$

and $g(x) = 2^x$

then $f(x) = x^2$

Example 1) let $F(x) = \sqrt{\tan x}$,

find f and g such that

$$F(x) = (f \circ g)(x)$$

Solution:

$$g(x) = \tan x$$

$$f(x) = \sqrt{x}$$

There are two functions involved here $\sqrt{x}$ and $\tan x$

check

$$(f \circ g)(x) = f(g(x))$$

$$= f(\tan x)$$

$$= \sqrt{\tan x} = F(x)$$

2) Let $F(x) = \cos \sqrt{x}$, Find f and

g such that $F(x) = (f \circ g)(x)$

Solution: $g(x) = \sqrt{x}$, $f(x) = \cos x$

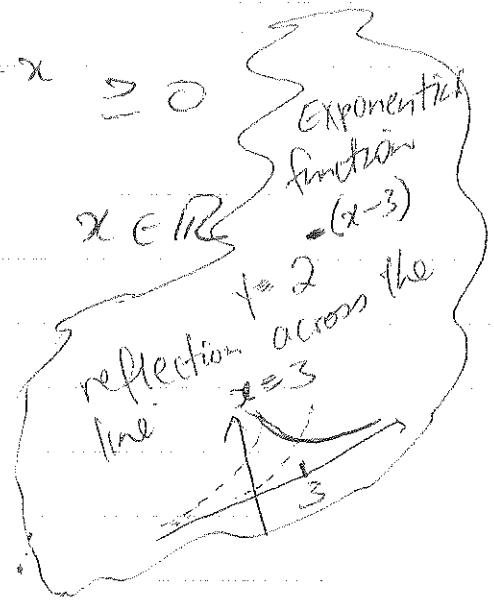
check
 $(f \circ g)(x)$

b) Find domain of $\sqrt{2^{3-x}}$

Solution: $\sqrt{2^{3-x}} \Rightarrow 2^{3-x} \geq 0$

But $2^{3-x} \geq 0$ for all $x \in \mathbb{R}$

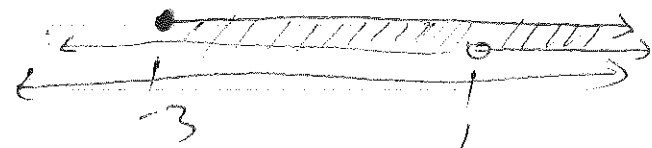
$\Rightarrow$ Domain is $\mathbb{R}$.



c) Domain of $\frac{\sqrt{x+3}}{x-1}$

Solution: $x+3 \geq 0$ and $x-1 \neq 0$

$\Rightarrow x \geq -3$ and $x \neq 1$



Domain is $\left\{ x / -3 \leq x < 1 \text{ OR } x > 1 \right\}$

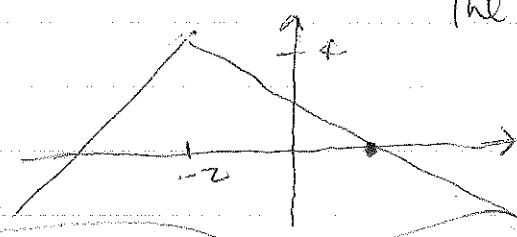
$= [-3, 1) \cup (1, \infty)$

Note: $y = -x$, reflection across y-axis, line $x=0$

$y = 2^{-x}$ " " "

$y = 2^{-(x-3)}$ " " " line $x=3$

$y = -|x+2|+4$, reflection across the line $y=4$



Composition of three functions

$$(f \circ g \circ h)(x) = f(g(h(x))) \\ = f(g(h(x)))$$

Example : If $f(x) = \frac{x}{x+1}$, $g(x) = \sin x$, $h(x) = x^2 + 1$

$$\text{then } (f \circ g \circ h)(x) = f(g(h(x))) \\ = f(g(x^2 + 1)) \\ = f(\sin(x^2 + 1)) \\ = \frac{\sin(x^2 + 1)}{\sin(x^2 + 1) + 1}$$

Example : If $(f \circ g \circ h)(x) = \cos^2(\sqrt{x})$

Find $f(x)$, $g(x)$ and $h(x)$

Solution :

$$h(x) = \sqrt{x}$$

$$g(x) = \cos x$$

$$f(x) = x^2$$

Check :

Hint ! : The formula for f says to first take the square root, then get the cos.

and then square, hence $F = \cos^2 x$ is such that

$$(f \circ g \circ h)(x) = (f \circ h)(x) \quad \text{where}$$

$h(x) = \sqrt{x}$; (first function is the square root)

$g(x) = \cos x$; (second function is the cos function)

$f(x) = x^2$; (third function is the quadratic function)

Example: If $F(x) = \sqrt{2^{3-x}}$, find

function f, g, h such that

$$F = f \circ g \circ h$$

Solution: $h(x) = 3-x$

$$g(x) = 2^x$$

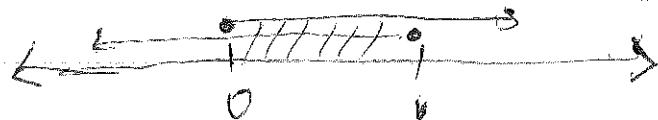
$$f(x) = \sqrt{x}$$

Exercises: (1) Find the domain of the following functions

a) $f(x) = \sqrt{x} + \sqrt{1-x}$

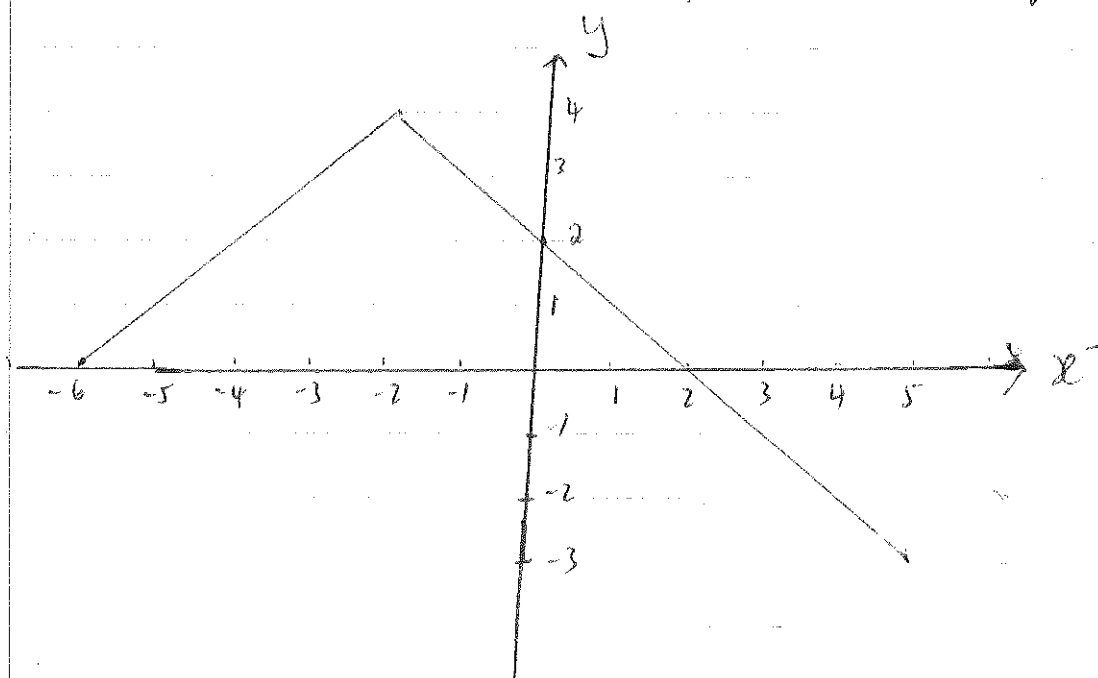
Solution: $x \geq 0$ and $1-x \geq 0$

$$\Rightarrow x \geq 0 \quad \text{and} \quad x \leq 1$$



Domain is $\{x \mid 0 \leq x \leq 1\} = [0, 1]$

Example : The following function is a transformation of $y = |x|$. Determine the function's equation.



The graph of $y = |x|$ has been shifted left 2 units, shifted up 4 units and reflected across the x -axis.

$$f(x) = -|x+2| + 4$$